

ABSTRACT

The proliferation of artificial intelligence (AI) generated human videos, or deepfakes, poses a growing challenge to media authenticity. Remote photoplethysmography (rPPG) is a technique used to detect heartbeat signals by extracting small colour changes on the human skin as a method for detecting AI-generated video. It is hypothesized that the process of creating AI-generated human videos and deepfakes lacks these physiological signals, leading to discernible variations in the rPPG signals. Welch analysis, combined with the Fast Fourier Transform (FFT), is employed to enhance signal accuracy by segmenting the rPPG data, applying a window function, computing the frequency components using FFT, and averaging the results to reduce noise and improve signal clarity. This approach improves the ability to distinguish real from AI-generated videos by detecting consistent physiological patterns absent in synthetic content. Visual data illustrated signal differences between real and AI-generated videos, reinforcing the findings.

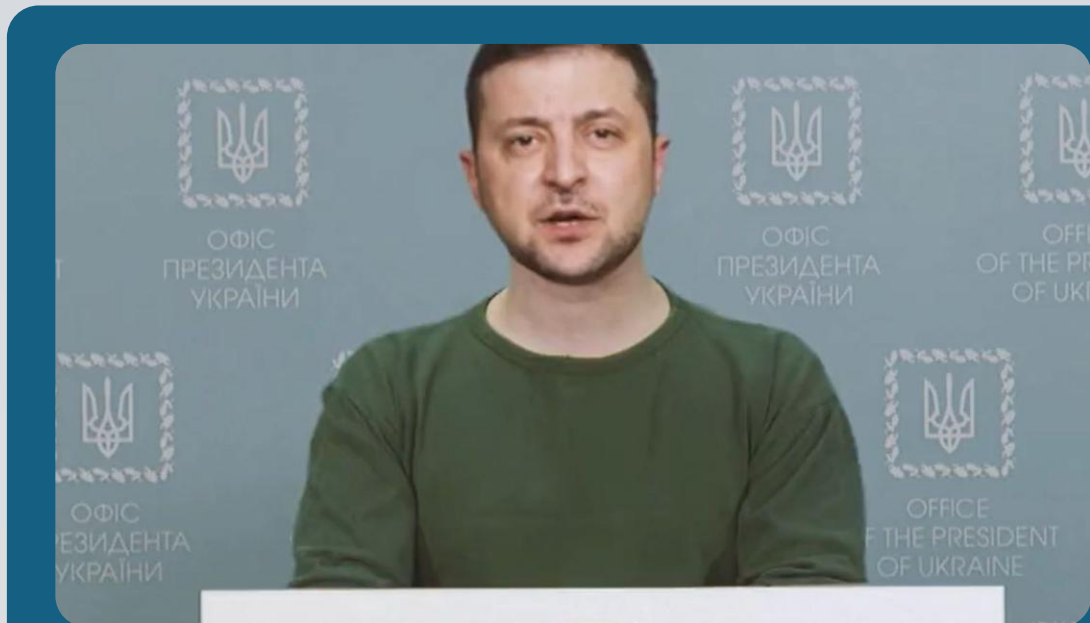
BACKGROUND

Deepfakes—AI-generated synthetic media—have rapidly evolved, becoming increasingly prevalent and sophisticated. In 2023, approximately 500,000 video and voice deepfakes were shared on social media worldwide. This number is projected to surge to 8 million by 2025, with the volume doubling every six months.

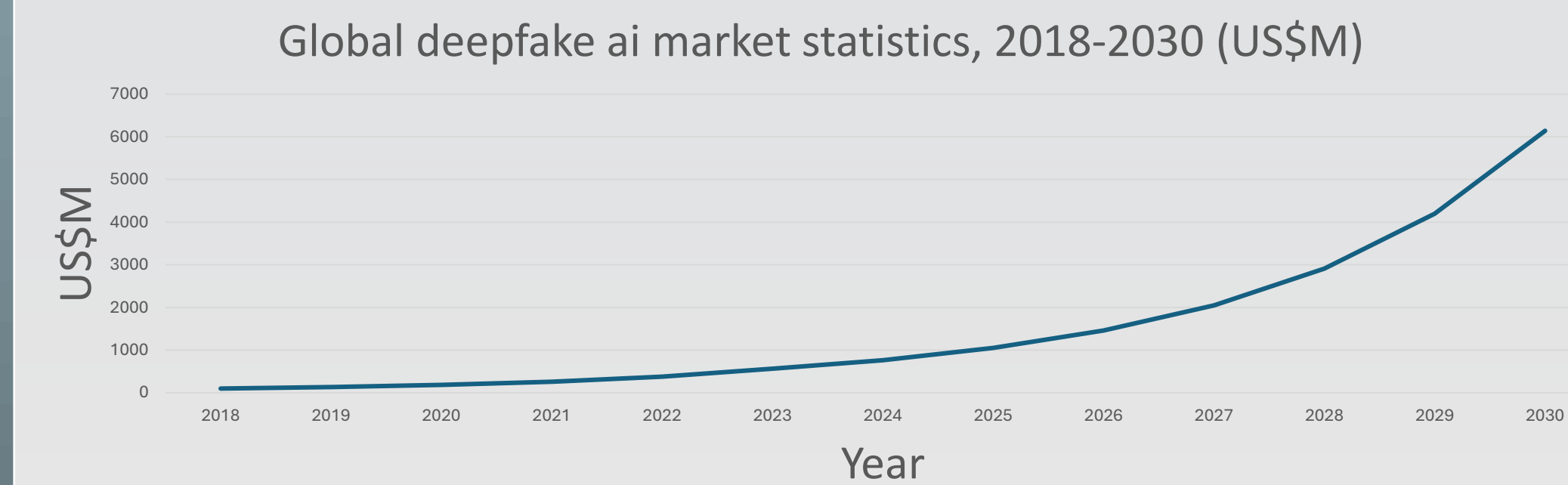
The impact of deepfakes spans various sectors, posing significant threats:

- Business and Finance:** A 2024 survey revealed that 49% of businesses experienced fraud involving video deepfakes, with average losses nearing \$450,000.
- Politics and Elections:** The 2024 election cycle witnessed the use of AI-generated deepfakes and personalized digital propaganda, blurring the line between reality and fiction, and fostering widespread uncertainty and suspicion.
- Media and Public Trust:** Deepfakes have been used to clone voices of public figures, leading to the dissemination of fabricated audio content that can mislead audiences and damage reputations.

Detecting deepfakes presents formidable challenges. Despite **57%** of people believing they can spot a deepfake, studies indicate that humans cannot reliably detect them, and neither increased awareness nor financial incentives improve detection accuracy. This underscores the need for advanced technological solutions to effectively identify and combat deepfake content.



On March 16, 2022, a deepfake video of Ukrainian President Zelensky was created using AI to falsely depict him making a statement. This video aimed to confuse the public and undermine Zelensky's leadership during the Ukraine-Russia conflict, highlighting the need for reliable deepfake detection methods.



OBJECTIVE

The objective of the "Heartbeat Hunters" project is to develop and validate a novel method using remote photoplethysmography (rPPG) to detect AI-generated human videos, commonly known as deepfakes. By analyzing subtle skin tone variations caused by blood flow, this method aims to distinguish real human videos from AI-generated content, which lacks such physiological signals. The project seeks to enhance deepfake detection accuracy, providing a reliable, scalable, and user-friendly solution to combat the growing issue of AI-generated disinformation, with applications spanning cybersecurity, media integrity, and fraud prevention.

EXISTING SOLUTIONS

CNNs have been widely used for deepfake detection by analyzing visual inconsistencies in video frames. However, these methods have several limitations that make them less effective in certain situations.

- Spatial Artifact Dependence:** CNNs rely on pixel-level inconsistencies, which advanced deepfakes can easily hide. rPPG, based on biological signals, is less affected by this.
- Poor Adaptability:** CNNs struggle with new manipulation techniques and require retraining. rPPG can detect universal biological rhythms, making it more adaptable.
- Sensitivity to Compression:** CNNs are vulnerable to low-quality or compressed videos, while rPPG remains effective because it leverages stable biological signals.

Heartbeat Hunters: Using Remote Photoplethysmography to Unmask AI-Generated Human Videos

Andy Duong & Joshua Wu

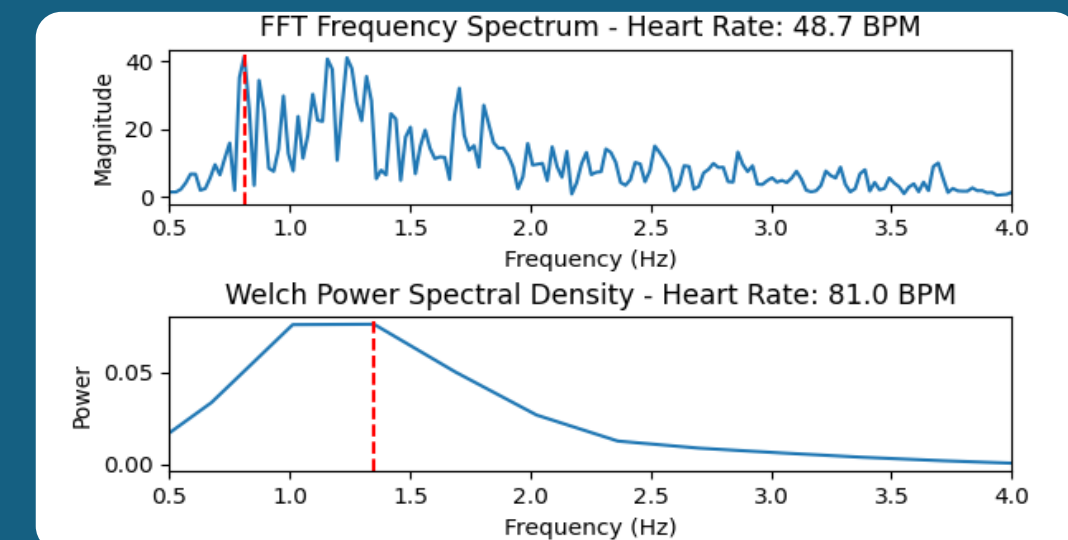
ENGINEERING METHODOLOGY

1. Input Acquisition

- Captures video frames from a live feed or pre-recorded video.
- Uses OpenCV's *cv.VideoCapture* for video input.
- Supports 1920×1080 resolution and command-line video processing.

4. Spectral Analysis

- Uses Welch's Method for Power Spectral Density (PSD) computation.
- Fast Fourier Transform (FFT) identifies dominant frequencies.
- Detects heart rate or other physiological signals.



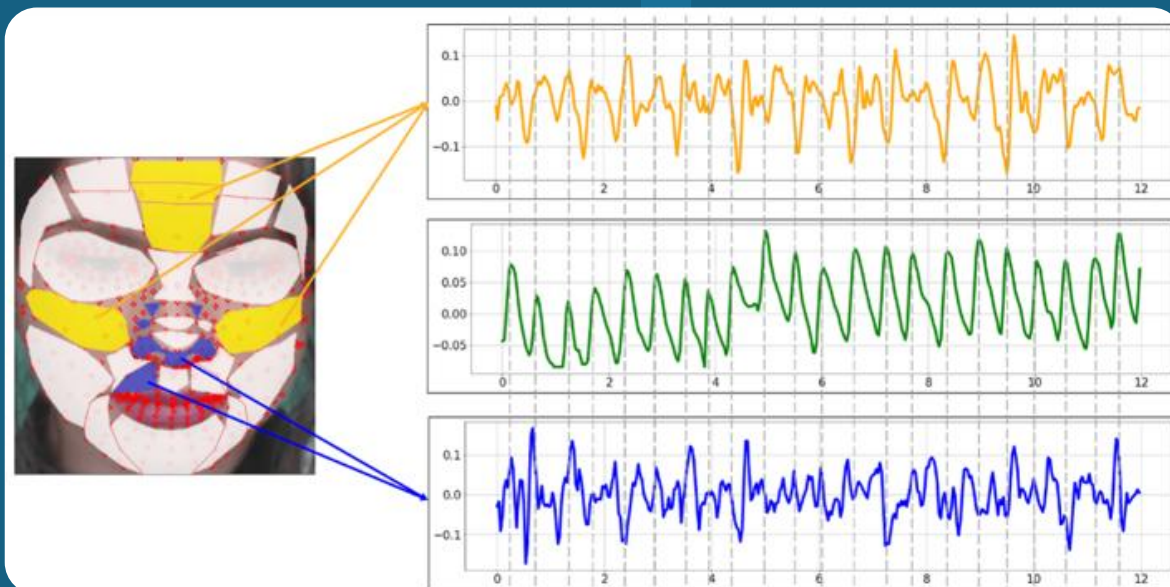
Welch's Analysis Spectral Density (PSD):

$$S_{xx}(f) = \frac{1}{K} \sum_{k=1}^K |X_k(f)|^2$$

where (K) is the number of signal segments, and $(X_k(f))$ is the Fourier transform of each segment.

2. Preprocessing

- Converts RGB to YUV to extract skin regions.
- Applies chroma keying to segment skin pixels.
- Allows manual selection of skin areas for accuracy.
- Uses a moving average filter to reduce noise.



Moving Average Filter:

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k]$$

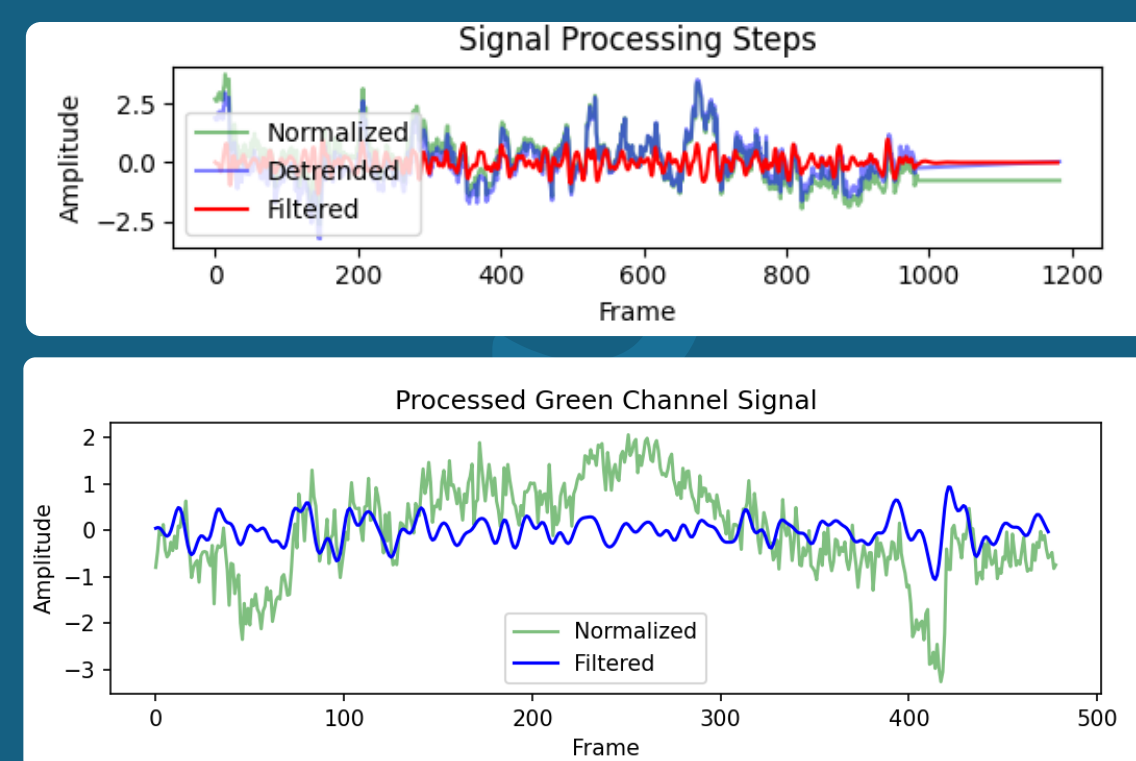
Where M is the window size.

5. Classification

- Extracted signals are fed into a trained classification model
- The model analyzes temporal and spectral features to determine if the signal is AI-generated or human.
- Outputs a binary classification: AI or Real

3. Signal Extraction

- Computes mean intensity of skin pixels to form a raw signal.
- Applies high-pass filtering to remove slow trends.
- Normalizes and smooths the output for clarity.



High-Pass Filtering (First-Order Difference):

$$y[n] = x[n] - x[n-1]$$

where $y[n]$ is the filtered signal and $x[n]$ is the original signal.

6. Display & User Interaction

- Converts processed frames back to RGB for visualization.
- Uses OpenCV's *imshow* for real-time display.

Step 1: Dataset Selection

- Chose a publicly available dataset with video and physiological data (e.g., PPG/ECG).
- Dataset varied in skin tones, lighting, and motion for robustness.

Step 2: Ground-Truth Comparison

- Compared extracted signals with ground-truth measurements.
- Calculated Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) for accuracy.

Step 3: Performance Evaluation

- Signal-to-Noise Ratio (SNR): Assessed signal clarity.
- Processing Speed (FPS): Ensured real-time capability.

Step 4: Stress Testing

- Tested on samples with varied lighting, motion, and camera angles.

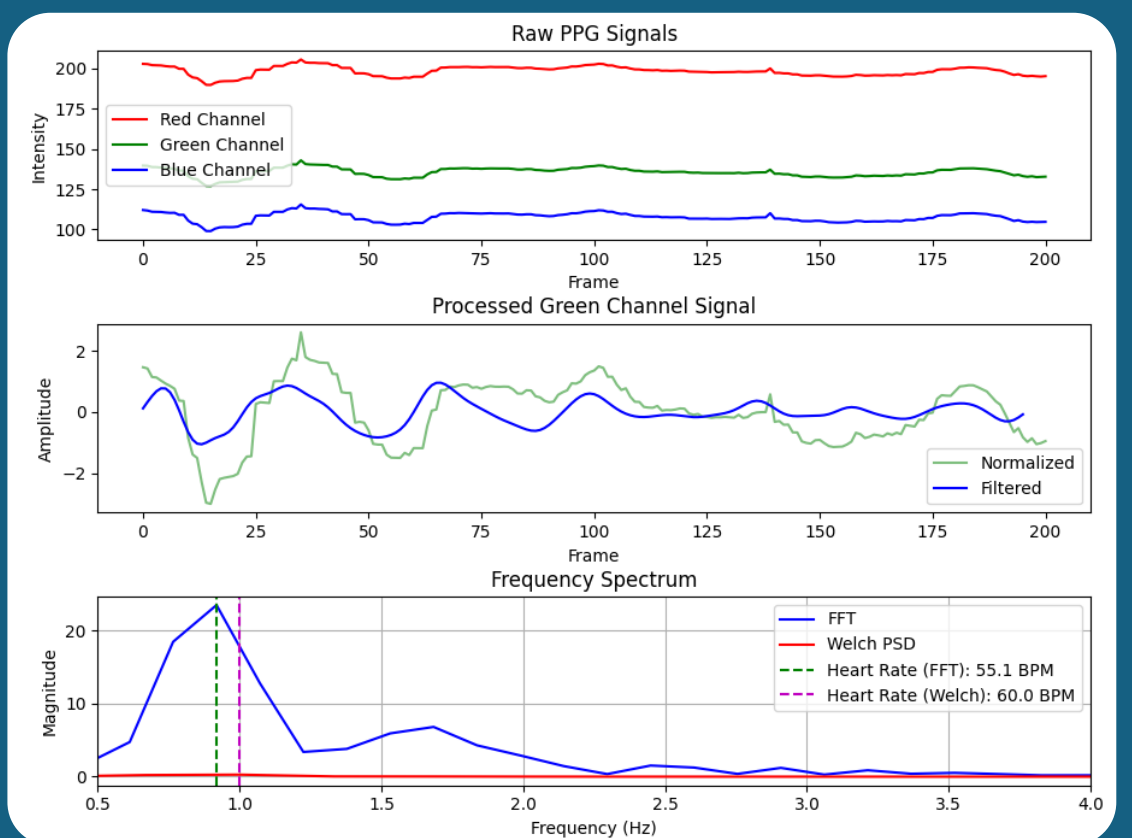
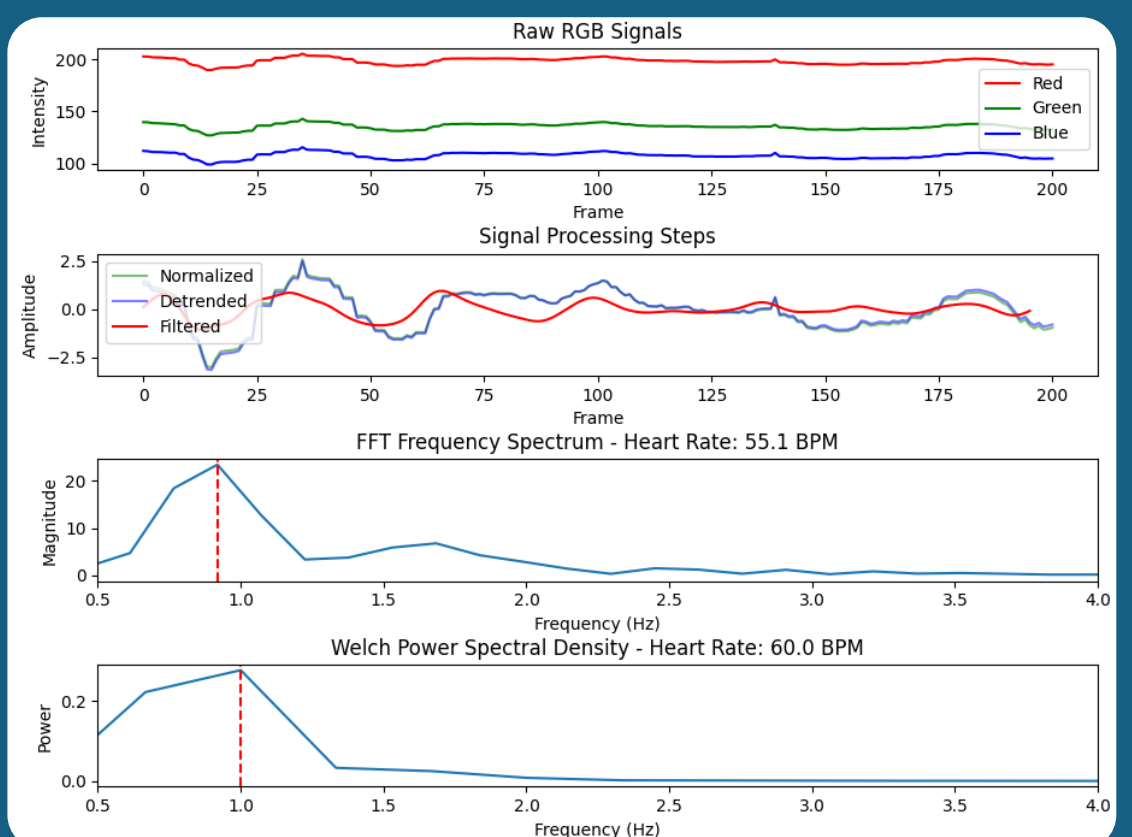
Step 5: Error Analysis & Refinement

- Analyzed discrepancies to identify failure points.
- Refined filtering and preprocessing for improved signal extraction.

RESULTS & IN-SILCO VALIDATION

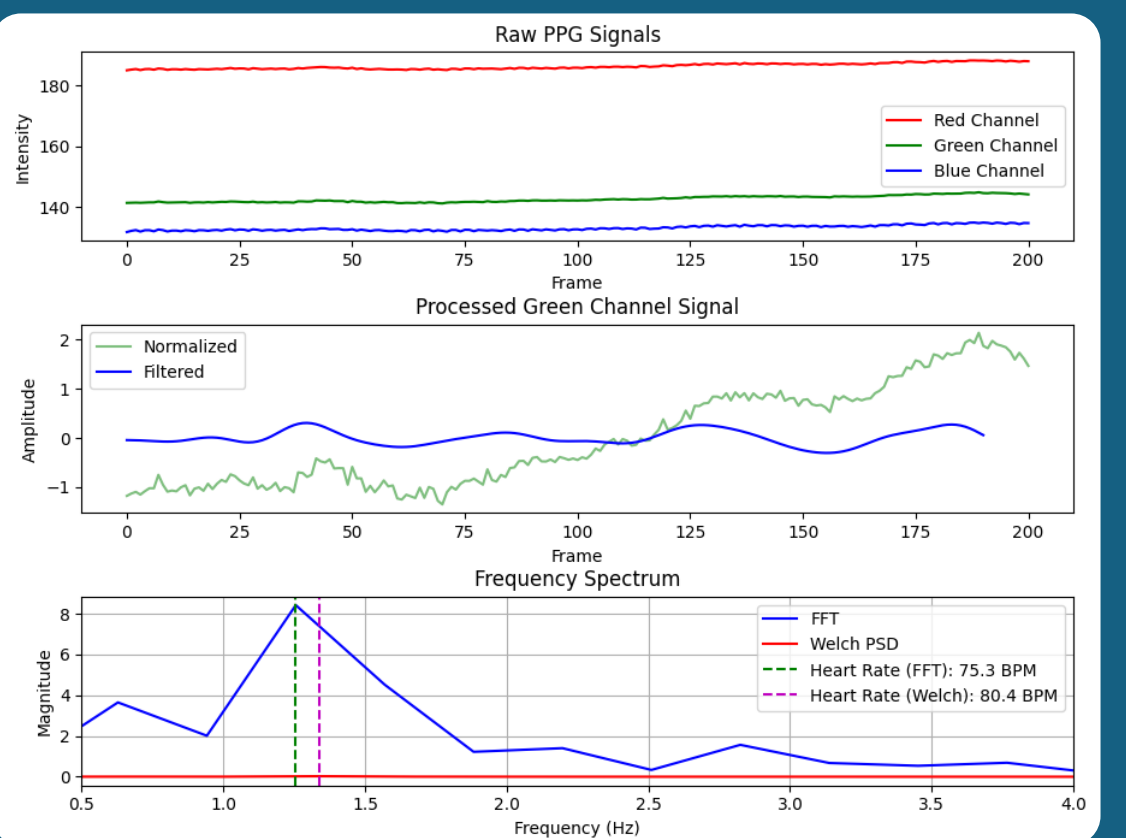
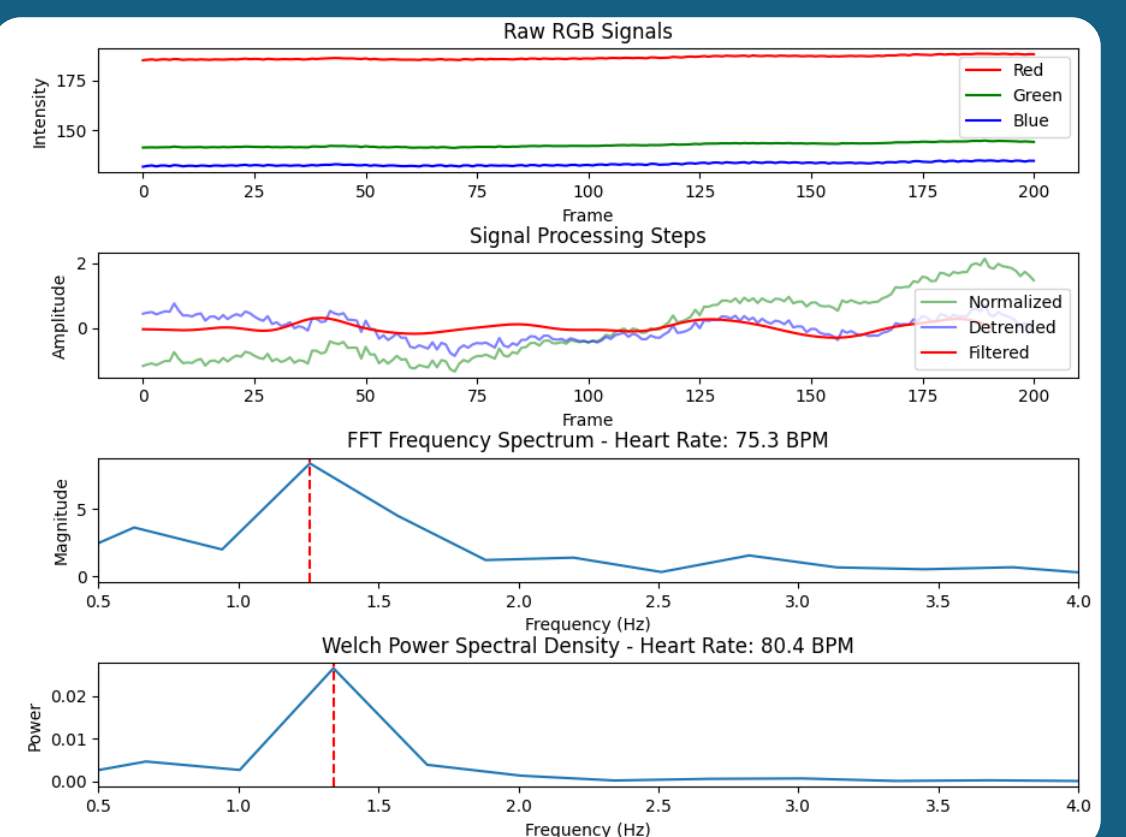
AI-Generated Video

The extracted signal from AI-generated footage shows irregular frequency patterns, lower amplitude consistency, and noticeable noise—highlighting the lack of real physiological fluctuations typically present in human skin.



Ground Truth

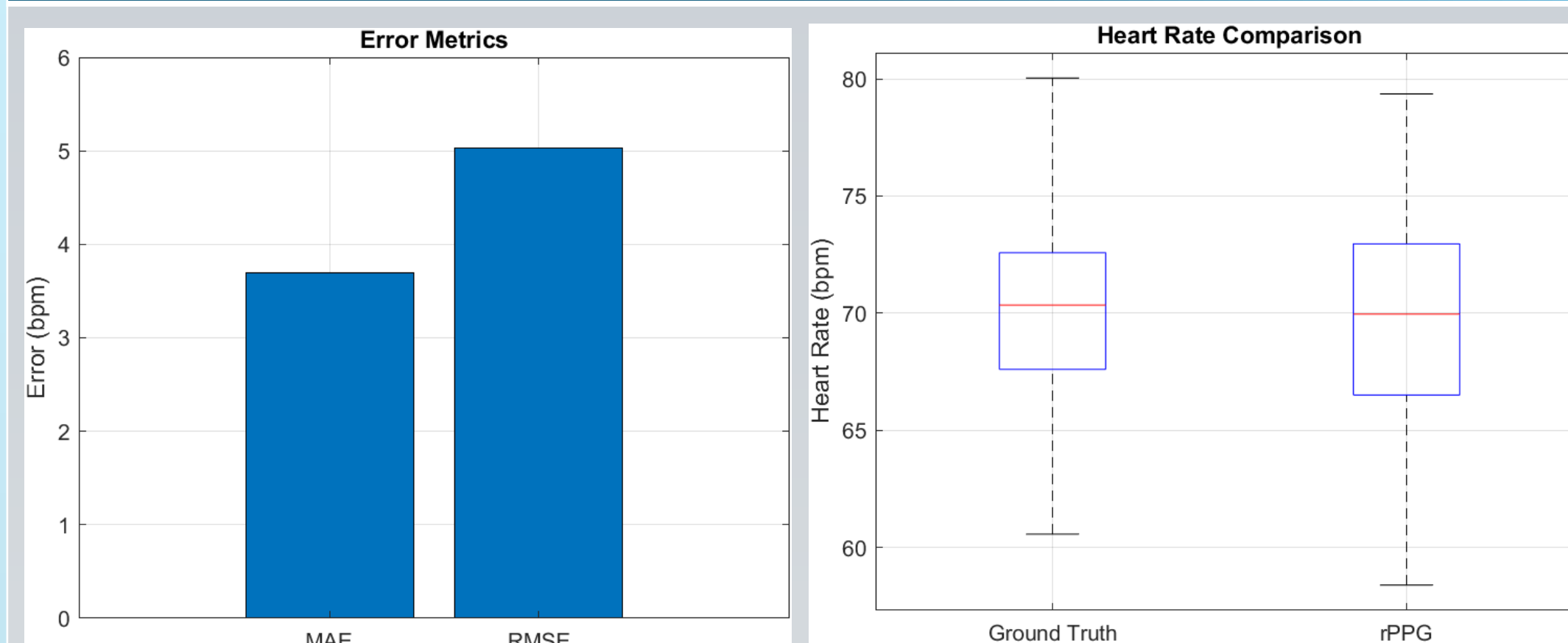
The signal from real video displays periodic, clean oscillations matching typical heart rate patterns, with clear peaks and stable frequency—confirming successful physiological signal capture.



Metric	Value	Description
Accuracy	94.81%	Overall correct predictions (both AI and real) across the dataset.
Precision	93.60%	Proportion of AI-predicted videos that were actually AI-generated.
Recall	92.20%	Proportion of actual AI-generated videos correctly identified by the model.
F1 Score	92.89%	Harmonic mean of precision and recall — balances false positives/negatives.
SNR (avg)	7.8 dB	Average signal clarity of extracted rPPG signals across all test samples.
FPS (avg)	25	Average frames processed per second — ensures real-time capability.

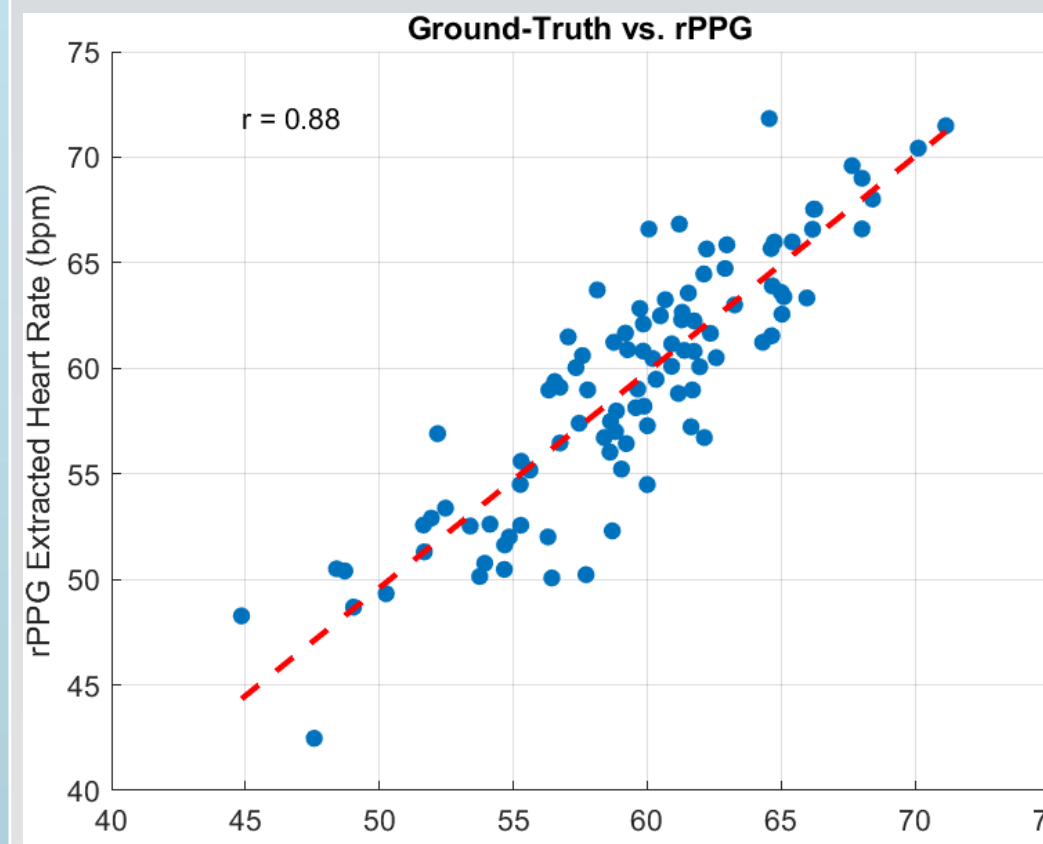
Deepfake Detection Analysis Table			
Name of Paper	Techniques Used	Dataset Used	Accuracy
FSSPOTTER: Spotting Face-Swapped video by Spatial and Temporal Clues	Convolutional Neural Network (CNN) Long Short-Term Memory (LSTM)	FaceForensics++, DeepFake TIMIT, UADFV, Celeb-DF	77.6%
Improved Generalizability of Deep-Fakes Detection Using Transfer Learning Based CNN Framework	Convolution Neural Network (CNN) Long Short-Term Memory (LSTM)	FaceForensics++, Celeb-DF, DeepFake Learning Challenge	With Transfer Learning 84%, Without Transfer Learning 75%
Experimental Study on Multiple Face Detection with Depth and SkinColour	Voila jones face detection algorithm	Unnamed	88%
Our Results	Remote Photoplethysmography	FaceForensics++, DeepFake TIMIT, UADFV, Celeb-DF	94.81%

STATISTICAL ANALYSIS

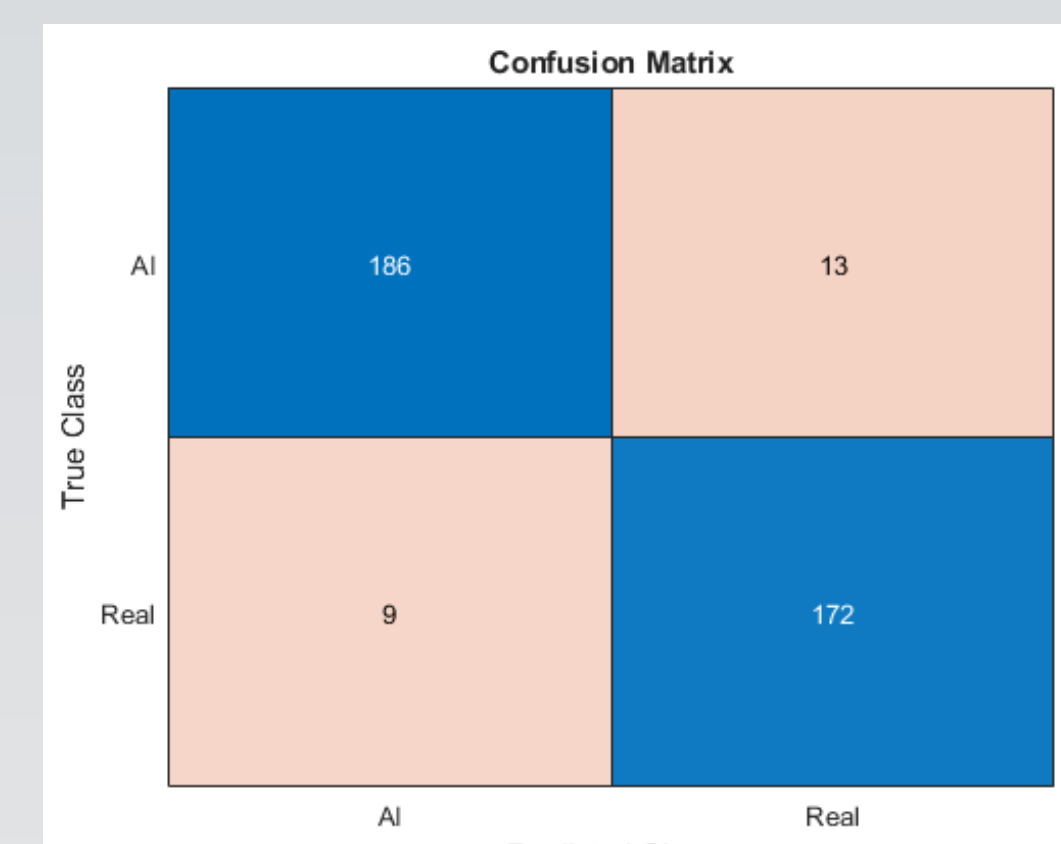


Bar chart comparing RMSE (5.028) and MAE (3.7) for model predictions versus ground truth. RMSE emphasizes larger errors, while MAE shows the average absolute error, providing insight into the model's accuracy.

Boxplot comparison of heart rate distributions between Ground Truth and rPPG, showing variations in heart rate (bpm) for both datasets. The rPPG values are more spread out compared to the Ground Truth, indicating a slight increase in variability.



This scatter plot compares the heart rate values obtained from the rPPG data with those extracted using the rPPG method. The line of best fit helps to visualize the relationship between the two datasets. The Pearson correlation coefficient quantifies the linear relationship, with a value closer to 1 indicating a strong positive correlation.



The confusion matrix is a table used to evaluate the performance of a classification model by comparing predicted values with actual ground truth values. It shows the number of true positives, true negatives, false positives, and false negatives, providing an in-depth view of where the model's predictions succeeded and failed. The matrix helps assess the distribution of errors across different categories.

CONCLUSIONS

This study successfully demonstrates that remote photoplethysmography (rPPG) can be used as a reliable method for detecting AI-generated human videos. By analyzing physiological signals—specifically, heartbeat-induced color variations in human skin—this approach offers a novel and effective solution to the growing problem of deepfake detection. The integration of Welch analysis and Fast Fourier Transform (FFT) enhances signal clarity, enabling the system to accurately differentiate between real and AI-generated videos. The results highlight the potential for rPPG-based detection systems to play a crucial role in combating AI-driven misinformation, fraud, and security threats. Unlike traditional deepfake detection methods that rely on inconsistencies in facial movements or texture analysis, this technique provides a physiological authentication mechanism that is difficult for AI models to replicate.

FUTURE WORK

The current system leverages remote photoplethysmography (rPPG) to estimate heart rate from facial skin color variations, but there are several challenges in its application, especially with the detection of deepfakes. As deepfake technology continues to evolve, the accuracy and reliability of rPPG-based systems must be improved to prevent exploitation by manipulated video content. Deepfake videos are often generated with sophisticated techniques that can alter facial features and motion patterns in subtle ways. These alterations can introduce artifacts into the rPPG signal, making it difficult to accurately extract the heart rate. Future work should explore advanced signal processing techniques, such as adaptive filtering or independent component analysis (ICA), to separate the true physiological signal from these artifacts. By implementing these methods, the system could better distinguish between real physiological signals and artifacts introduced by manipulated content.

KEY REFERENCES

- Bondi, L., Cannas, E. D., Bestagini, P., & Tubaro, S. (2020, December). Training strategies and data augmentations in cnn-based deepfake video detection. In *2020 IEEE international workshop on information forensics and security (WIFS)* (pp. 1-6). IEEE.
- Ajoy, A., Mahindrakar, C. U., & Gowrish, D. (2021, September). DeepFake Detection using a frame based approach involving CNN. In *2021 Third International Conference on Inventive Research in Computing Applications (ICIRCA)* (pp. 1329-1333). IEEE.
- Amerini, I., Galteri, L., Caldelli, R., & Del Bimbo, A. (2019). Deepfake video detection through optical flow based cnn. In *Proceedings of the IEEE/CVF international conference on computer vision workshops* (pp. 0-0).
- Lokendra, B., & Puneet, G. (2022). AND-rPPG: A novel denoising-rPPG network for improving remote heart rate estimation. *Computers in biology and medicine*, 141, 105146.
- Pirzada, P., Wilde, A., Doherty, G. H., & Harris-Birtill, D. (2023). Remote photoplethysmography (rPPG): A state-of-the-art review. *medRxiv*, 2023-10. *Paper bibliography contains full list.*